



Sampling Distributions II

Megan Ayers
Stat 141 | Fall 2026
Friday, Week 6

Goals for Today

- Discuss the framework for random sampling
- Investigate properties of the “sampling distribution”

The ♥ of statistical inference is quantifying uncertainty

Like with regression, need to distinguish between the **population** and the **sample**

Parameters:

- Based on the **population**
- Unknown if we don't have data on the whole population
- EX: β_0 and β_1
- EX: μ (population mean) p (population proportion)

Statistics:

- Based on the **sample** data
- Known (calculated from our data)
- Usually used to estimate a population parameter
- EX: $\hat{\beta}_0$ and $\hat{\beta}_1$
- EX: $\bar{x}$ (sample mean), $\hat{p}$ (sample proportion)

Estimation and Uncertainty

- We are interested in the value of a **parameter** in a *population*, and use a **statistic** from a *sample* to estimate the parameter.
 - e.g., want to know the proportion (p) of Reed students with October birthdays
 - Estimate p by using the proportion in a sample, ($\hat{p}$), of 100 individuals.
- **Key question:** How accurate is $\hat{p}$ as an estimate of p ?
- **Sub-question:** If we take many samples, how much will $\hat{p}$ vary?
- The distribution of all possible values of $\hat{p}$ (for a fixed sample size) is called the **Sampling Distribution of $\hat{p}$**

Sampling Distribution of a Statistic

Steps to Construct a Sampling Distribution:

1. Decide on a sample size, n .
2. Determine all* possible samples of size n from the population.
3. Compute the sample statistic in each sample.

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Approximate Sampling Distribution of a Statistic

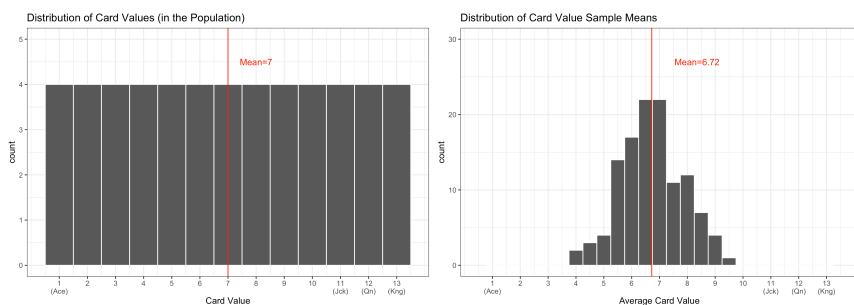
Steps to Construct an **Approximate** Sampling Distribution:

1. Decide on a sample size, n .
2. **Randomly select a sample of size n (without replacement) from the population.**
3. Compute the sample statistic in that sample.
4. **Put the whole sample back in.**
5. Repeat Steps 2 - 4 many (1000+) times.

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Population and Sampling Distributions in Our Activity

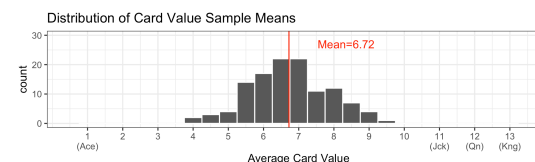
Last Class: got a “snapshot” of the sampling distribution for the **mean card value from a sample of 10 cards**.



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Sampling Distribution in Our Activity

“Snapshot” of sampling distribution for **mean card value from a sample of 10 cards**:



Why just a “snapshot”?

- **Q:** How many elements are there in the sampling distribution?

$$\binom{52}{10} = 15,820,024,220 \text{ Samples}$$

- We only took about 100 samples (combining both sections)...

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Sampling Distribution Creation: a smaller scale

Consider a **population of 4 students**, where the (true) population proportion of October birthdays is $p = \frac{1}{4} = 0.25$

name	bday_month
Joel	october
Maria	january
Arthur	april
Klaus	september

- **Practice:** With a neighbor, write out the sampling distribution for $\hat{p}$ (the sample proportion of those with October birthdays) with...
 1. All possible samples of size 2
 2. All possible samples of size 3
- And, what are the means of these two sampling distributions?

05:00

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Sampling Distribution Creation: a smaller scale

1. Sampling distribution for the sample proportion $\hat{p}$ from a sample of 2:

name	bday_month
Joel	october
Maria	january
Arthur	april
Klaus	september

sample	phat
Joel + Maria	0.5
Joel + Arthur	0.5
Joel + Klaus	0.5
Maria + Arthur	0.0
Maria + Klaus	0.0
Arthur + Klaus	0.0

- There are $\binom{4}{2} = 6$ elements in the sampling distribution
- **Mean=0.25**, which is p !

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Sampling Distribution Creation: a smaller scale

2. Sampling distribution for the sample proportion $\hat{p}$ from a sample of 3:

name	bday_month
Joel	october
Maria	january
Arthur	april
Klaus	september

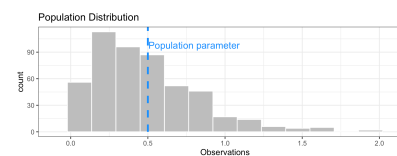
sample	phat
Joel + Maria + Arthur	1/3
Joel + Maria + Klaus	1/3
Joel + Arthur + Klaus	1/3
Maria + Arthur + Klaus	0

- There are $\binom{4}{3} = 4$ elements in the sampling distribution
- **Mean=0.25**, which is p ! (again)
- **Note:** The means of the sampling distributions were all $p = \frac{1}{4}$ *even though none of the sample proportions $\hat{p}$ were equal to $\frac{1}{4}$!*

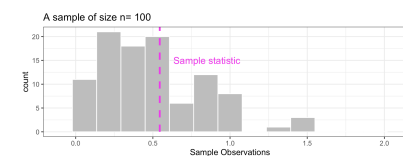
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Pause to Check In: Three Different Distributions

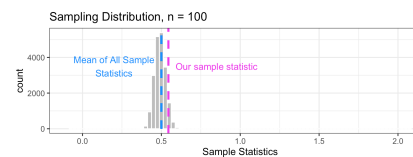
What we want to know:



What we usually have:



Sampling distribution of the statistic:



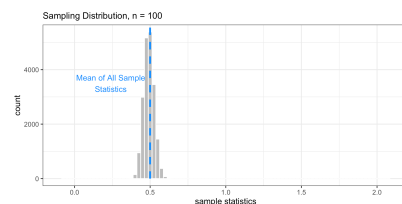
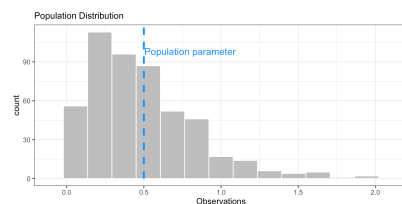
Discuss with your neighbor: What makes these distributions related, but different? Think back to the context of the card sampling activity from Wednesday as an example.

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Sampling Distribution vs. Population Distribution

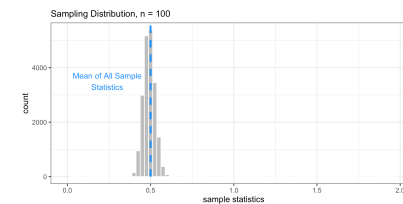
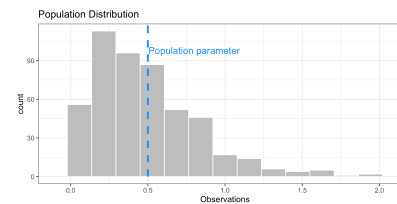
- Q: How do the centers compare?
- Q: How does the spread compare?



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Sampling Distribution Shape

- Q: How do the shapes compare?



- For most sample statistics and sufficiently large sample sizes ($n \geq 30$ is a rule of thumb), **the sampling distribution will be approximately bell-shaped**, even if the population is not.

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Sampling Distribution: Standard Error

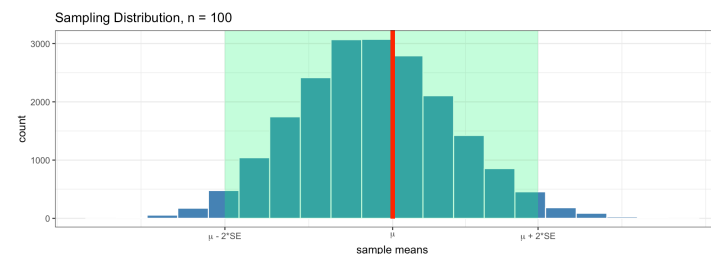
- Just like any distribution, the sampling distribution of a statistic has a mean, standard deviation, median, IQR, etc.
 - We give its standard deviation a special name...
 - ... the **Standard Error**: standard deviation of a *sampling distribution* (i.e. of a *statistic*)
 - Different from the standard deviation of *observations* in the sample or in the population!
- Ex. When estimating a population proportion (p) with the sample proportion ($\hat{p}$) with sample size n , theory shows that:
 - The sampling distribution for $\hat{p}$ has mean p
 - and has standard deviation (aka standard error) $SE = \sqrt{\frac{p(1-p)}{n}}$
- This means if the population proportion $p = 0.20$ and $n = 100$, then in the sampling distribution:
 - $\text{mean}(\hat{p}) = p = 0.20$
 - $SE = \sqrt{\frac{0.20(1-0.20)}{100}} = 0.04$

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Characterizing Sampling Variability

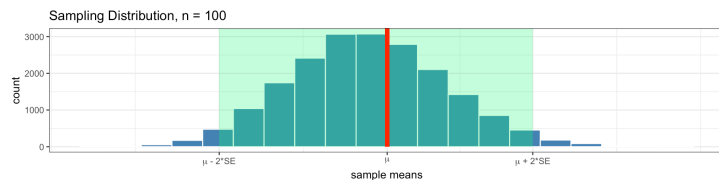
Remember that the **standard error (SE)** of a sample statistic measures variability between different samples.

- For **$\approx$ bell-shaped distributions**, about 95% of observations fall within **two standard errors** of the population's mean μ .
- **Very useful implication!**
 - The sampling distribution for the sample average, $\bar{x}$ is $\approx$ bell-shaped
 - ... and is centered at the population mean, μ .
- So **95% of all sample statistics, $\bar{x}$, fall within 2 standard errors of μ !**



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This is very powerful!



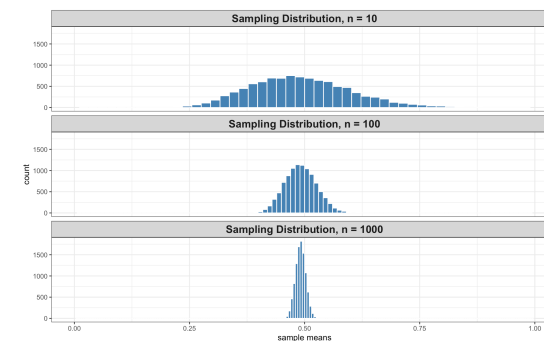
Even though we just have one sample, if we know SE , we know how far away from the parameter our statistic is likely to fall!

- This allows us to give quantitative plausible ranges for the parameter
- We only need SE to infer this, not the whole sampling distribution

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Variability and Sample Size

Variability of the sampling distribution generally decreases as sample size increases

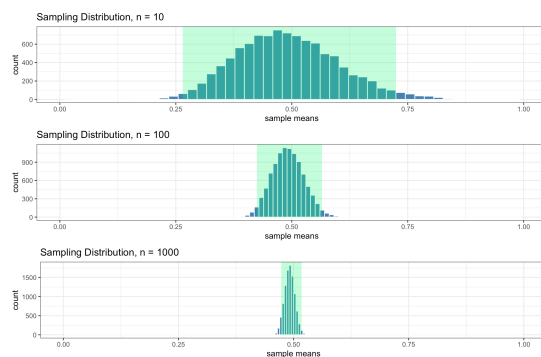


- **Implication:** Our statistic $\bar{x}$ is likely to be closer to parameter μ as sample size n increases.

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Variability and Sample Size

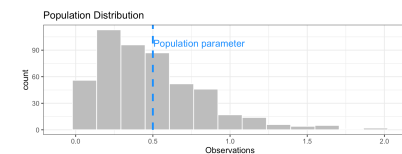
- These sampling distributions for $n = 10$, 100 , and 1000 are all approximately bell-shaped, and so 95% of sample means are within 2 standard errors of the population mean.
- Highlighted in green are the intervals containing 95% of all sample means:



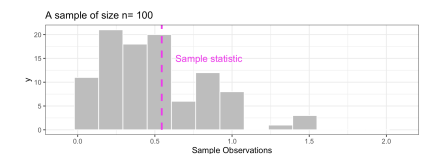
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How do Sampling Distributions Help Us?

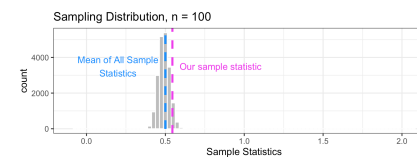
What we want to know:



What we have:



What sampling distributions tell us about what we have:



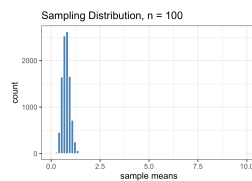
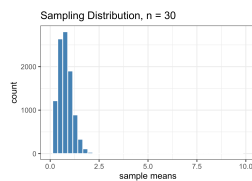
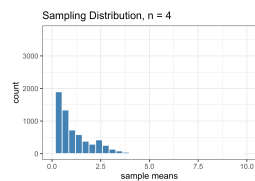
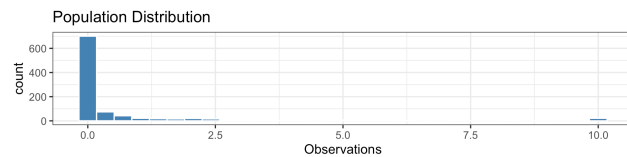
- If we had **just the standard error** of the sampling distribution, we could get a sense of how far off any single statistic is likely to be from the parameter.

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The Shape of the Sampling Distribution: word of warning

When $n \geq 30$, the sampling distribution is *usually* bell-shaped. But sometimes, you need a large sample size.

- Example: A funky population distribution.



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Recap

Key Features of a Sampling Distribution

- Centered around the true population parameter.
- As the sample size increases, the **standard error** (SE) of the statistic decreases.
- As the sample size increases, the shape of the sampling distribution becomes more bell-shaped and symmetric.

Q: If I am estimating a parameter in a real example, why won't I be able to construct the sampling distribution?

- **Big takeaway:** We (often) don't need to know the entire sampling distribution to assess uncertainty, SE would be enough
- **Remaining problem?** Okay, but SE is still a population parameter!
- **Solution:** We can **estimate** SE using our sample data

Next time:

- Using simulation to learn about the shape and SE of the sampling distribution from a single sample

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